CRISP-DM: Data Understanding

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In this report, I present key findings in the selected Steam dataset 2025 and visually explain the data.

1 Collect Initial Data

1.1 Data Requirements and Selection Criteria

The target dataset is Steam Dataset 2025: Multi-Modal Gaming Analytics available on Kaggle. The dataset contains information on applications published on Steam in the last 30 years. Since the business focuses on finding game developers, all other types of applications will be excluded. Moreover, seeking high relevance in the market, the analysis should be based on the last 10 years (2015-2025) of the game market.

1.2 Selected Data

1. `applications`: app metadata
2. `reviews`: user reviews (sampled)
3. `developers`: developer entities
4. `publishers`: publisher entities
5. `genres`: application genres
6. `categories`: application categories
7. `application_developers`: game-developer relationships
8. `application_publishers`: game-publisher relationships
9. `application_genres`: Links apps to genres.
10. `application_categories`: Links apps to Steam categories

2 Describe Data

Since most of the data are concentrated in tables `applications` and `reviews`, a description analysis is performed on them.

2.1 Applications Table (Games Only, 2015-2025)

After filtering for type='game' and release year between 2015 and 2025, the applications table contains $n = 150,279$ games. Table 1 shows the proportion of non-null values for each field.

Table 1: Game Data Availability (n=158,215)

Field	Availability	Notes
appid, name, type, is_free, release_date	1.00	Core identifiers – complete
required_age, background, header_image	1.00	Metadata – complete
mat_supports_linux/windows/mac	1.00	Platform support – complete
created_at, updated_at	1.00	Timestamps – complete
category_name, genre_name	1.00	Classification – complete
short_description	1.00	Text description – complete
supported_languages	1.00	Language support – complete
mat_pc_os_min	0.98	Minimum OS requirements
mat_pc_memory_min	0.95	Minimum RAM requirements
mat_pc_processor_min	0.94	Minimum CPU requirements
mat_pc_graphics_min	0.88	Minimum GPU requirements
mat_currency	0.84	Price currency
mat_initial_price	0.84	Initial price (cents)
mat_final_price	0.84	Final price after discount
mat_discount_percent	0.84	Discount percentage
mat_achievement_count	0.66	Number of achievements – 34% missing
mat_pc_os_rec	0.55	Recommended OS – 45% missing
mat_pc_memory_rec	0.53	Recommended RAM – 47% missing
mat_pc_processor_rec	0.52	Recommended CPU – 48% missing
mat_pc_graphics_rec	0.51	Recommended GPU – 49% missing
recommendations_total	0.23	Total positive votes – 77% missing
metacritic_score	0.04	Metacritic score – 96% missing

Fields usable for analysis:

- `mat_initial_price`, `mat_final_price`, `mat_discount_percent` – pricing strategy (84% available)
- `mat_achievement_count` – game depth (66% available – use with caution)
- `mat_supports_windows/mac/linux` – platform support (complete)
- `genre_name`, `category_name` – content classification (complete)
- `supported_languages` – localization effort (complete)
- `release_date` – temporal analysis (complete)
- `recommendations_total` – too sparse (23% available) – exclude from modeling
- `metacritic_score` – too sparse (4% available) – exclude

2.2 Reviews Table

The `reviews` table contains sampled user reviews (n = 1,048,148), with a maximum of 10 reviews per application. Table 2 describes the key fields and their availability.

Table 2: Reviews Table Field Description (n=1,048,148)

Field	Non-null Count	Description / Notes
recommendationid	1,048,148	Primary key – unique review ID
appid	1,048,148	Foreign key to applications
author_steamid	1,048,148	Reviewer Steam ID
author_num_games_owned	1,048,148	Games in reviewer's library (complete)
author_num_reviews	1,048,148	Reviews written by reviewer (complete)
author_playtime_forever	1,048,145	Total playtime (minutes) – 3 missing (< 0.1%)
author_playtime_last_two_weeks	1,048,145	Recent playtime – 3 missing
author_last_played	1,048,145	Timestamp of last play – 3 missing
language	1,048,148	Review language (complete)
review_text	1,047,369	Review content – 779 missing (0.1%)
timestamp_created/updated	1,048,148	Timestamps (complete)
voted_up	1,048,148	Positive recommendation (complete)
votes_up	1,048,148	Helpful vote count (complete)
weighted_vote_score	1,048,148	Steam's helpfulness score (0.0-1.0) – complete
steam_purchase	1,048,148	Purchased on Steam (complete)
received_for_free	1,048,148	Free acquisition (complete)
written_during_early_access	1,048,148	Early access review (complete)

Fields usable for analysis:

- `author_playtime_forever`, `author_playtime_last_two_weeks` – player engagement (> 99.9% available)
- `author_last_played` – recency of play (> 99.9% available)
- `author_num_games_owned`, `author_num_reviews` – reviewer experience (complete)
- `voted_up` – sentiment (complete)
- `weighted_vote_score` – perceived helpfulness (complete)
- `review_text` – available for NLP (99.9% available)
- `language` – review language distribution (complete)
- `author_playtime_at_review` – playtime at time of review (83% available – use with caution)

Note: The dataset samples only 10 reviews per application, which limits the depth of analysis but provides broader coverage across games compared to the sparse `recommendations_total` field.

3 Explore Data

3.1 Business Objective 1

The following hypotheses were formulated to address the problem of finding qualities of developers that work with top publishers:

1. What p should be selected to cover a substantial volume on Steam (> 50% with the limit on number of top publishers not exceeding 30% of major publishers)?
2. How are publishers and developers represented?
3. How useful are reviews in terms of describing a developer?
4. What application features are insightful about a developer?

Filtering Data

The selection criteria were applied to identify the representation of the game market in the dataset (Figure 1).

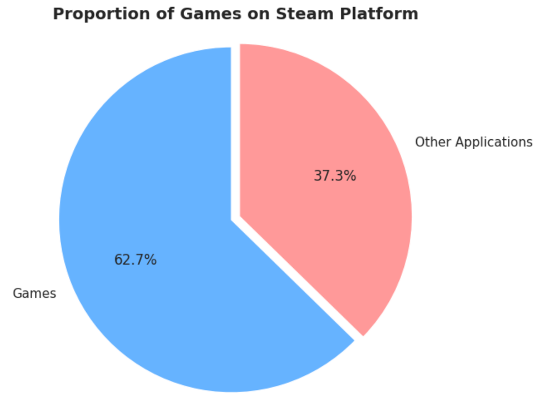
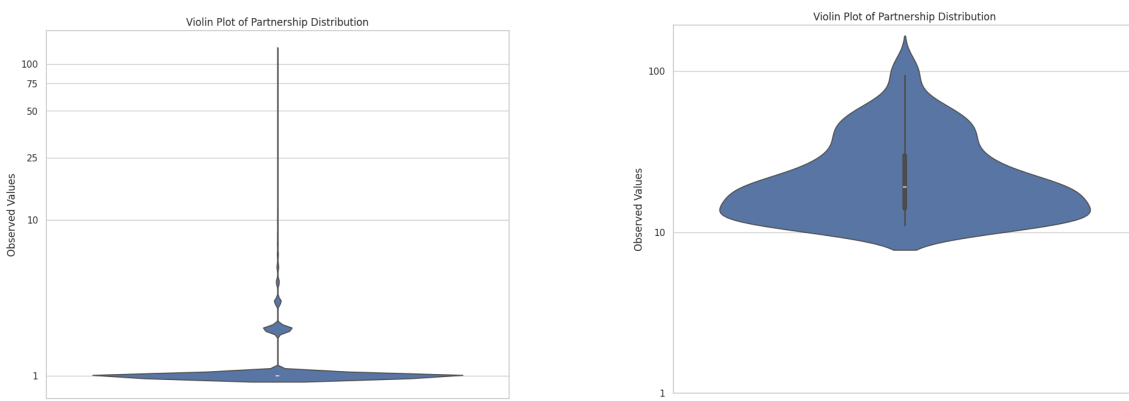


Figure 1: Game Proportions Chart

Define Top Publishers

Figure 2 showcases that selecting publishers purely based on volume is not sufficient: almost 90% of publishers presented have only 1 partnership (in fact, these publishers are indie developers in 92.7% of cases). Therefore, only publishers with more than 10 partnerships were considered as being in the top. These publishers are called *major*.



(a) No filtration on partnership count

(b) Partnerships > 10 for publishers

Figure 2: Differences in Partnership Distribution among Publishers

Finally, for the selected group of major publishers, Figure 3 displays the cumulative volume coverage of the games produced by these publishers (not of all market volume, to reduce noise from indie developers) and the cumulative partnerships coverage of all developers that worked with them (among all developers). In addition, the plot demonstrates the ranks of publishers with a substantial portion of the volume market (> 50%) that are in the first 30% of all ranks.

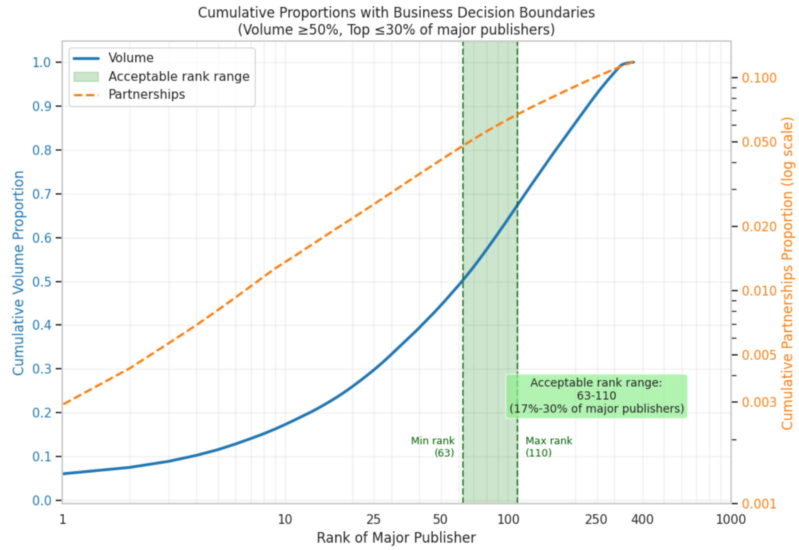


Figure 3: Volume-Partnerships Coverage for Major Publishers

Thus, the threshold of 100 was selected to determine the first 100 publishers as the top. Figure 4 demonstrates the distribution of developers in the specified setting.

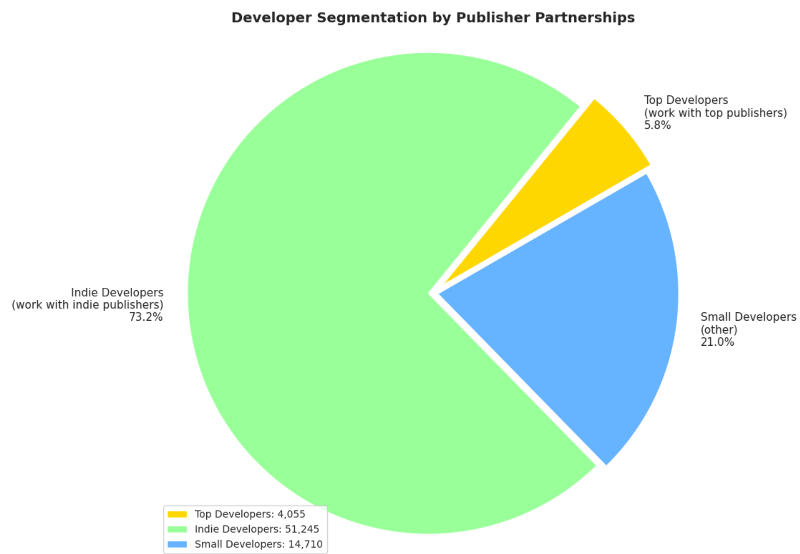


Figure 4: Developers Distribution after setting $p = 100$

Exploring Reviews

The two main sources of information about developers are games and their reviews. This section reports on how game reviews describe a developer.

First, the dataset contains the number of all reviews of a game in the column `recommendations_total` in the table `applications`. However, the majority games do not have any information in this field. In contrast, the dataset provides up to 10 sampled reviews per game, covering a much larger proportion of games. Manual aggregation of reviews results in a greater amount of information about a developer, making this feature more reliable compared to `recommendations_total` (Figure 5).

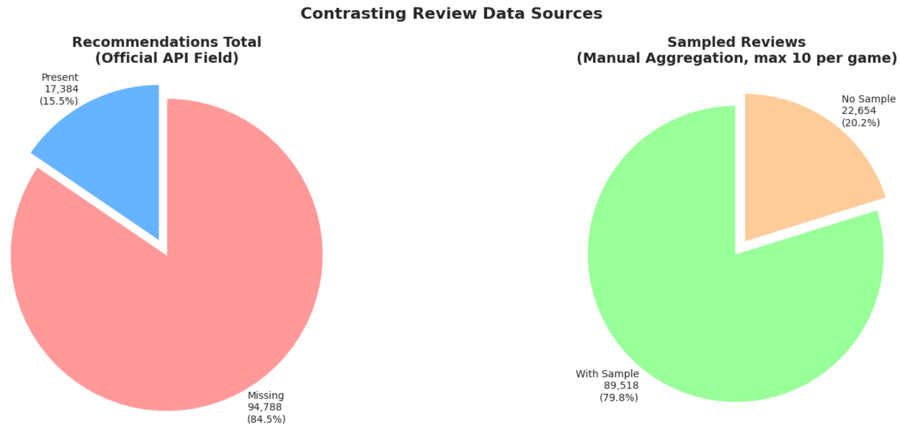


Figure 5: Reviews Availability

Next, the following metrics were identified and tested to distinguish between top developers and non-top developers using macro and micro aggregation of reviews:

Table 3: Review-based Developer Metrics Description

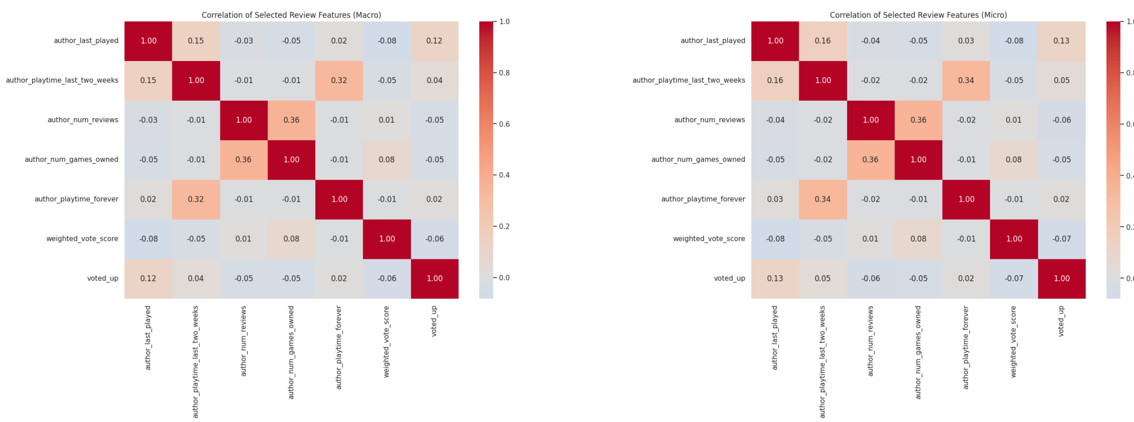
Metric	**Aggregation Type**	**Description**
`author_last_played`	Macro/Micro	Average time since players last played any game by this developer (days)
`author_playtime_last_two_weeks`	Macro/Micro	Average playtime in the last two weeks across all players of developer's games (minutes)
`author_num_reviews`	Macro/Micro	Average number of reviews written by players who reviewed this developer's games
`author_num_games_owned`	Macro/Micro	Average number of games owned by players who reviewed this developer's games
`author_playtime_forever`	Macro/Micro	Average total playtime across all players of developer's games (minutes)
`weighted_vote_score`	Macro/Micro	Steam's helpfulness score for reviews (0.0-1.0), averaged per developer
`voted_up`	Macro/Micro	Proportion of positive reviews (True/False ratio)

The differences in features between two groups (non-top developers, top developers) were tested using Welch's test with two-sided alternative (confidence level 5%):

Table 4: T-test Results: Top vs. Non-top Developers

Metric	Aggregation	p-value	Reject H_0 ($p < 0.025$)
author_last_played	Macro	< 0.001	Yes
author_playtime_last_two_weeks	Macro	< 0.001	Yes
author_num_reviews	Macro	< 0.001	Yes
author_num_games_owned	Macro	< 0.001	Yes
author_playtime_forever	Macro	< 0.001	Yes
weighted_vote_score	Macro	< 0.001	Yes
voted_up	Macro	0.71	No
author_last_played	Micro	< 0.001	Yes
author_playtime_last_two_weeks	Micro	< 0.001	Yes
author_num_reviews	Micro	< 0.001	Yes
author_num_games_owned	Micro	< 0.001	Yes
author_playtime_forever	Micro	< 0.001	Yes
weighted_vote_score	Micro	< 0.001	Yes
voted_up	Micro	0.69	No

Finally, the correlation between features is evaluated for both aggregations:



(a) Macro aggregation

(b) Micro aggregation

Figure 6: Feature Correlation Heatmaps: Macro vs. Micro Aggregation

Key Takeaways:

1. Developers who partner with top publishers produce games that:
   - have players with significant change in play time (`author_last_played`, `author_playtime_last_two_weeks`, `author_playtime_forever`);
   - also have players with differences in number of owned games (`author_num_games_owned`) and number of reviews (`author_num_reviews`);
   - receive different helpfulness scores (`weighted_vote_score_macro`).
2. Positive vote ratio (`voted_up`) is not correlated with group of developer.
3. Macro and micro averaging demonstrate the same result.

Exploring Games

Then, the game feature analysis is reported. The following metrics were proposed for evaluation:

Table 5: Game-based Developer Metrics Description

Metric	Type	Description
avg_initial_price	Continuous	Average initial price of developer's games (in cents)
percent_free	Proportion	Percentage of developer's games that are free-to-play
avg_extra_platform_support	Continuous	Average number of platforms supported beyond Windows (Mac, Linux)
avg_achievement_count	Continuous	Average number of achievements per game
genre_count	Discrete	Number of unique genres across developer's portfolio
avg_discount_percent	Continuous	Average discount percentage applied to games
category_count	Discrete	Number of unique Steam categories across developer's portfolio
years_past_first_release	Discrete	Years since developer's first game release (as of 2025)
release_frequency	Continuous	Average number of games released per year
avg_languages	Continuous	Average number of supported languages per game
max_languages	Discrete	Maximum number of languages supported in any single game

Similarly, the differences in features between groups were evaluated using Welch's test with two-sided alternative (confidence level 5%):

Table 6: T-test Results: Top vs. Non-top Developers (Game-based Metrics)

Metric	p-value	Reject H_0 ($p < 0.025$)
avg_initial_price	< 0.001	Yes
percent_free	< 0.001	Yes
avg_extra_platform_support	0.96	No
avg_achievement_count	< 0.001	Yes
genre_count	0.01	Yes
avg_discount_percent	< 0.001	Yes
category_count	< 0.001	Yes
years_past_first_release	< 0.001	Yes
release_frequency	< 0.001	Yes
avg_languages	< 0.001	Yes
max_languages	< 0.001	Yes

Likewise, the correlation analysis between features is presented in Figure 7.

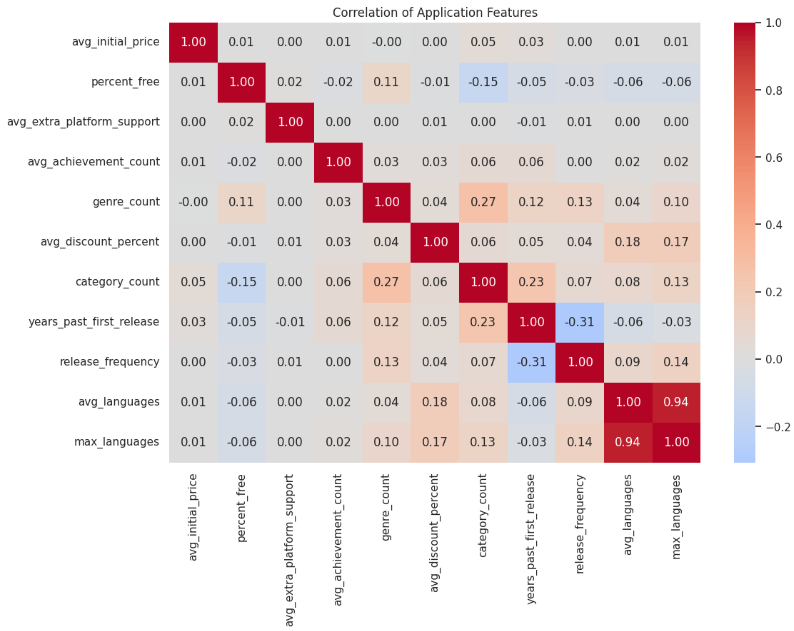


Figure 7: Game Feature Correlation Heatmap

Key Takeaways:

1. Developers who partner with top publishers produce games that:
   - have different mean starting price (`avg_initial_price`);
   - have different mean number of achievements (`avg_achievement_count`);
   - have different number of categories (`category_count`) as well as genres (`genre_count`);
   - differ in percent of free games (`percent_free`) and average discount (`avg_discount_percent`);
   - differ in frequency of releases and years from the first release (`release_frequency`, `years_past_first_release`);
   - differ in mean and max number of supported languages (`avg_languages`, `max_languages`).
2. Average extra platform support (`avg_extra_platform_support`) was not found to correlate with group of developer.
3. Maximum supported languages count (`max_languages`) does not itself carry too much information and can be replaced with `avg_languages` fully.

3.2 Business Objective 2

The second goal is related to ranking emerging developers. The key analysis was to identify the proportions of emerging developers among all developers and the proportions of top developers among emerging developers.

Figure 8 summarizes the findings:

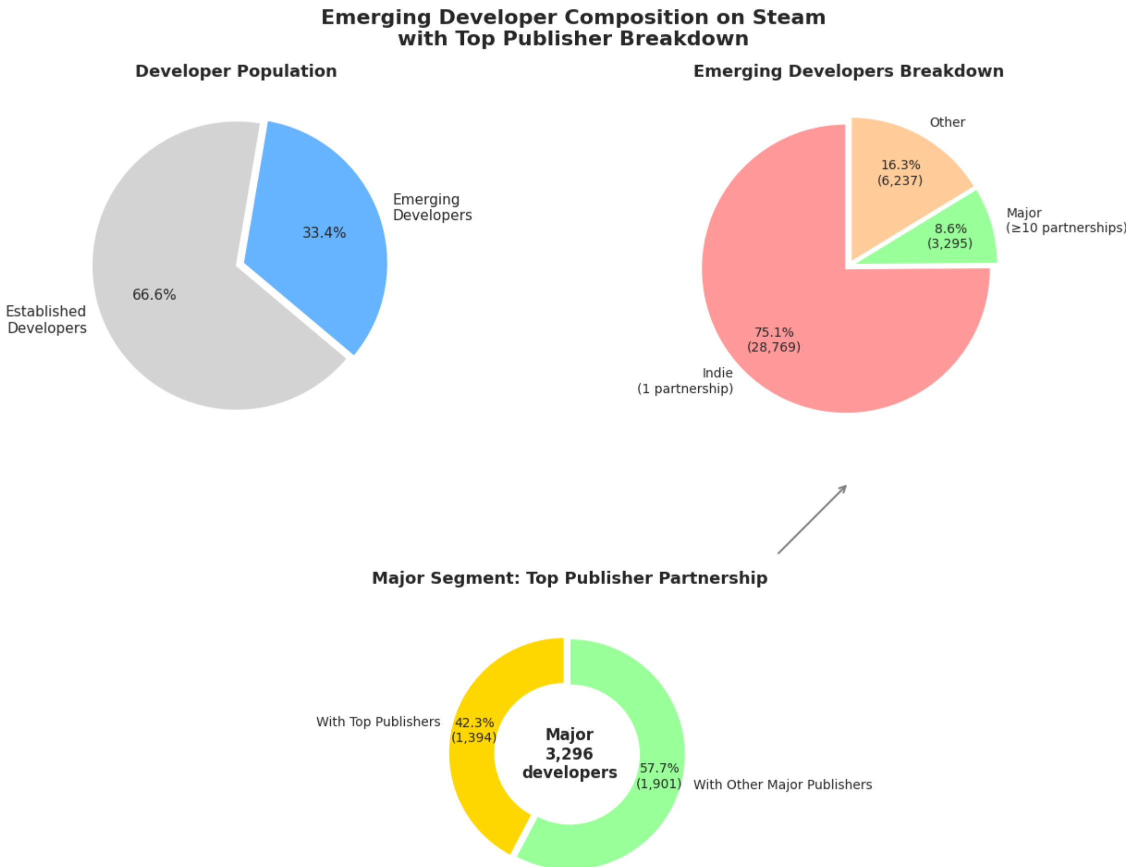


Figure 8: Emerging Developers Distribution

4 Verify Data Quality

4.1 Data Quality Assessment

During the data exploration, I systematically evaluated the quality of the fields that will be used for modeling. The main issues, together with the planned handling, are summarized in Table 7.

Table 7: Data quality issues and handling strategies

Issue	**Affected fields**	**Handling strategy**
Missing price information	`mat_initial_price`, `mat_final_price`, `mat_discount_percent`	Free games (91.4% of missing) imputed with 0; remaining missing imputed with median (robust).
Currency diversity	`mat_currency`	Non-USD prices converted to USD using approximate exchange rates.
Unrealistic price values	`mat_initial_price`	Verified via Steam API; genuine outliers retained as they carry signal.
Missing language data	`supported_languages`	Set to 1 (English only) for the few games with missing information.
Missing achievement count	`mat_achievement_count`	Imputed with median (robust); a binary flag indicates missingness.
Missing genre / category	`genre_name`, `category_name`	Counts set to 0 for developers with no assigned genres/categories.

These issues will be resolved during the Data Preparation phase. The handling choices are designed to preserve as much signal as possible while ensuring the data are suitable for modeling.

In the last part, the availability and completeness of the resulting data are reported. Data availability refers to developers aggregation information availability (e.g. having games or games with reviews). On the other hand, data completeness is related to presence of all features for a developer.

Figure 9 depicts the proportion of available game and review data:

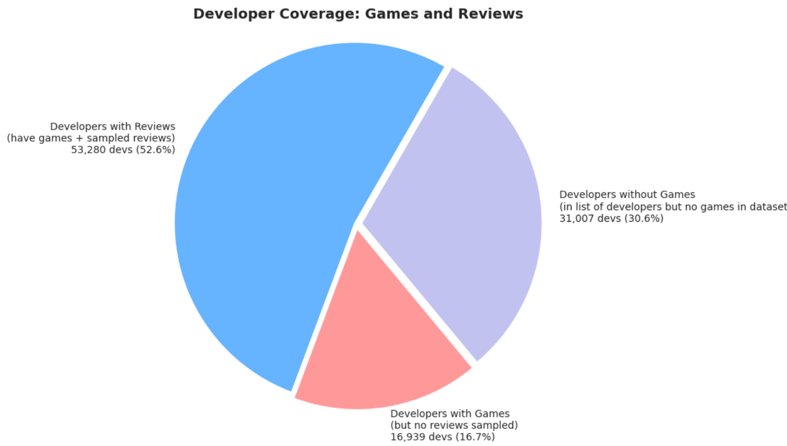


Figure 9: Review and Game Data Availability Chart

Developers without relevant game information do not bring any relevant information for this analysis. Moreover, around 17% of the developers with games lack review data, which limits the quality of inference. The distribution of top developers across the developers with games is as follows:

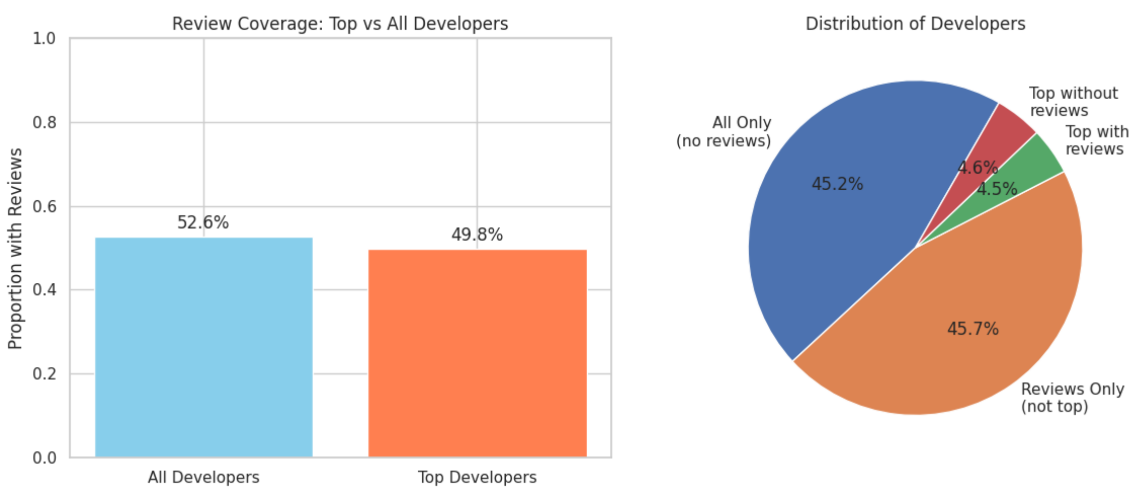


Figure 10: Review Availability among Top Developers

The developers with games can also be studied from the completeness point of view:

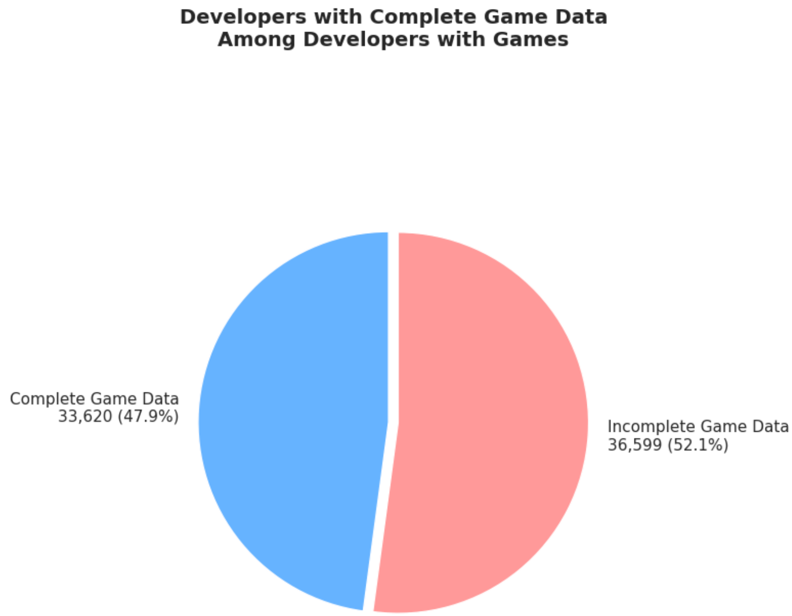


Figure 11: Game Information Completeness

Focusing on the last group of developers (having both games and reviews), Figure 12 highlights the distribution of top developers:

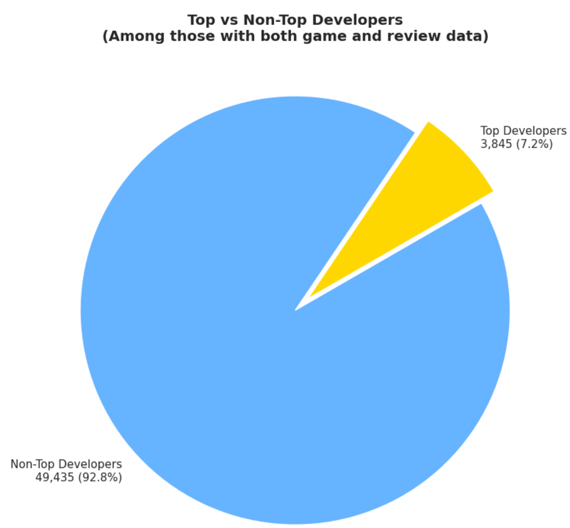


Figure 12: Top Developers Distribution among Game and Review Available Data

And with completeness taken into account:

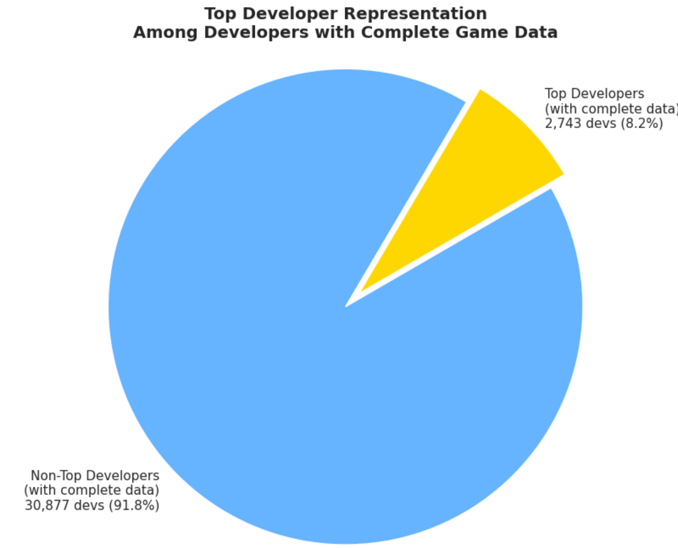


Figure 13: Top Developers Distribution among Game and Review Complete Data

Finally, the availability and completeness for emerging developers is summarized in Figure 14.

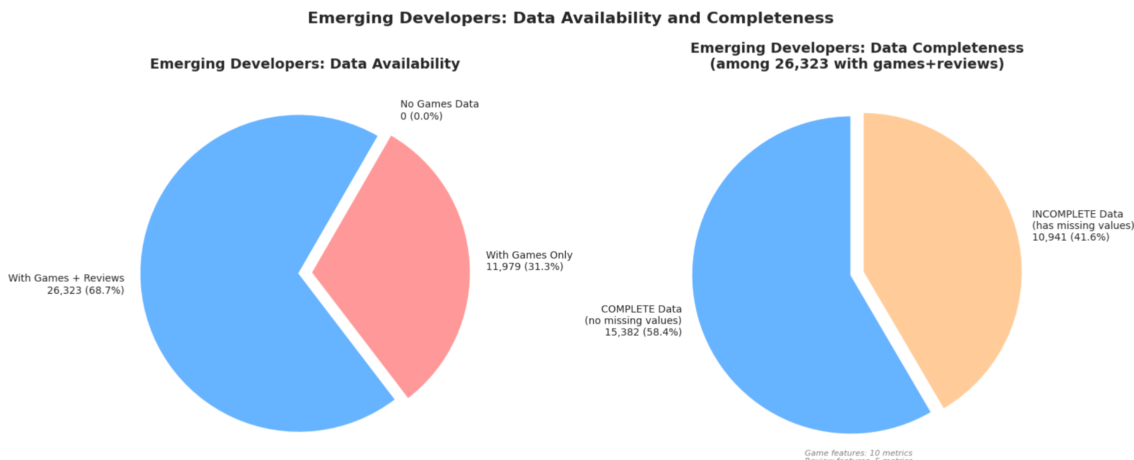


Figure 14: Distribution of Emerging Developers among Available and Complete Data